

Alexander Sludds

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WORK EXPERIENCE

PHOTONIC ARCHITECT, LIGHTMATTER COPORATION

March 2023 - Present | Boston, MA

Staff Photonic Architect

- Technical leadership: Lead a task force of 20-30 engineers to solve key challenges in simulating and closing link budgets for our products.
- Pioneered pathfinding for Lightmatter's 200G/400G optical modulators, establishing design roadmaps for next-gen products.
- Performed hardware analysis of the benefits of including semiconductor optical amplifiers (SOAs) in our DWDM link
- Created simulation methodologies to include accurate models of photonics, analog electronics and SERDES together to estimate full system performance
- Developed methodologies for system level crosstalk and power penalty (interchannel and intrachannel) and device specifications to meet those power penalties
- Developed methodologies for calculating TDECQ of transmitter modulator + driver in multiple simulation tools
- Utilized monte-carlo methodologies in simulation flow to predict product yield as well as inform post-silicon mitigation strategies
- Created tasks, worked with technical program managers to delegate tasks, vetted results and combined results
- Weekly readout on task force performance to senior leadership highlighting progress, successes and failures

Senior Photonic Architect

- Electronic-photonic co-design: Created models + modeling methodology of passive and active photonic IP in verilogA. Closely worked with analog team to enable co-simulation of photonic components and analog devices.
- Device design: Designed custom photonic sub-systems including vernier filters, broadband wavelength interleavers. Improved optical power handling of silicon thermal phase shifter to record breaking numbers (Watt class).
- Hardware bring-up: Led hardware demonstrations with customers of packaged photonic links.
- Created documentation framework and specifications for photonics tapeouts.

INTERN, LIGHTMATTER COPORATION

May 2022 - September 2022 | Boston, MA

RESEARCH ASSISTANT

MIT RLE: DIRK ENGLUND QUANTUM PHOTONICS

July 2018 - March 2023 | Cambridge, MA

- Published in high impact journals and conferences including Science, Nature Photonics, Science Advances, Physical Review X, CLEO, Photonics West and OFC.
- Led multiple silicon photonic tapeouts through a CMOS pilot line foundry (AIM photonics) incorporating designs with hundreds of active components that are used simultaneously.

EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT)

Cambridge, MA

Ph.D in Electrical Engineering (February 2023)

M.Eng. in Electrical Engineering and Computer Science (May 2019)

B.S. in Electrical Engineering and Computer Science (May 2018)

BRUNSWICK COMMUNITY COLLEGE (BCC)

Supply, NC | June 2014

Associates Degree in Arts

Associates Degree in Science

AWARDS

- NSF Graduate Research Fellowship (received April 2019)
- Best Paper Award at OECC/PSC 2022
- BCC President's Award : Highest GPA in class

PUBLICATIONS

- Delocalized Photonic Deep Learning on the Internet's Edge. A. Sludds, et al. Science 2022
- Deep Learning with Coherent VCSEL Neural Networks. Z. Chen, A. Sludds, et al. Nature Photonics 2023
- Large-Scale Optical Neural-Networks based on Photoelectric Multiplication. R. Hamerly, L. Bernstein, A. Sludds et al. Physical Review X (2020)
- Freely scalable and reconfigurable optical hardware for deep learning. L. Bernstein*, A. Sludds*, et al. Scientific Reports (2021)

SKILLS

SOFTWARE

Python • Numpy • Cadence Virtuoso • Keysight ADS • VPI Photonics • LaTeX • MATLAB • Lumerical (Mode, FDTD, Interconnect)

HARDWARE

Vector-Network Analyzer • Lightwave Component Analyzer • Optical Spectrum Analyzer • Bit Error Rate Tester • Wafer Testers

LICENSES

FCC Commercial Licenses (GROL, GMDSS Operator & Maintainer, Marine Radio, Ship Radar) • Amateur Radio Extra Class (KC1GAZ)

TEACHING EXPERIENCE

MIT

Jan 2016 - Dec 2022 | Cambridge, MA

- Instructor and TA for 8+ MIT EECS courses including Silicon Photonics, Control Theory, and Electromagnetics. Experience designing course material, leading labs, and lecturing.
- Independently taught "Introduction to Signals and Systems" during MIT's IAP for three consecutive years.